□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

2026 10

CPU

English Title: End-to-End Argument Mining via Relation-Aware Learning and Dependency-Closed Repair

NONE

CDCP

AAEC CDCP F1 73.29 52.02 8.10 13.26 3.63 4.89 T1 4.48 8.37 14 Holm CDCP

Abstract

End-to-end argument mining identifies text spans, argumentative roles, and directed relations. Component–relation dependencies create cascading effects during both local paraphrasing and prediction repair. This thesis investigates dependency maintenance in training and inference, building on multi-candidate reranking and bounded repair.

For training, relation influence domains identify internal, cross-boundary, and NONE factors. Lossless annotation transport and factor masks define reliable correspondences for auxiliary supervision and paired distribution alignment, while original documents retain full-graph generation targets. For inference, dependency-closed transactions jointly reconsider component changes and all affected relations under frozen full-object scores, separate utility gates, and edit budgets. Local and global optimization share the same object identity and constraints. CDCP collective-source relations are evaluated as complete native hyperedges.

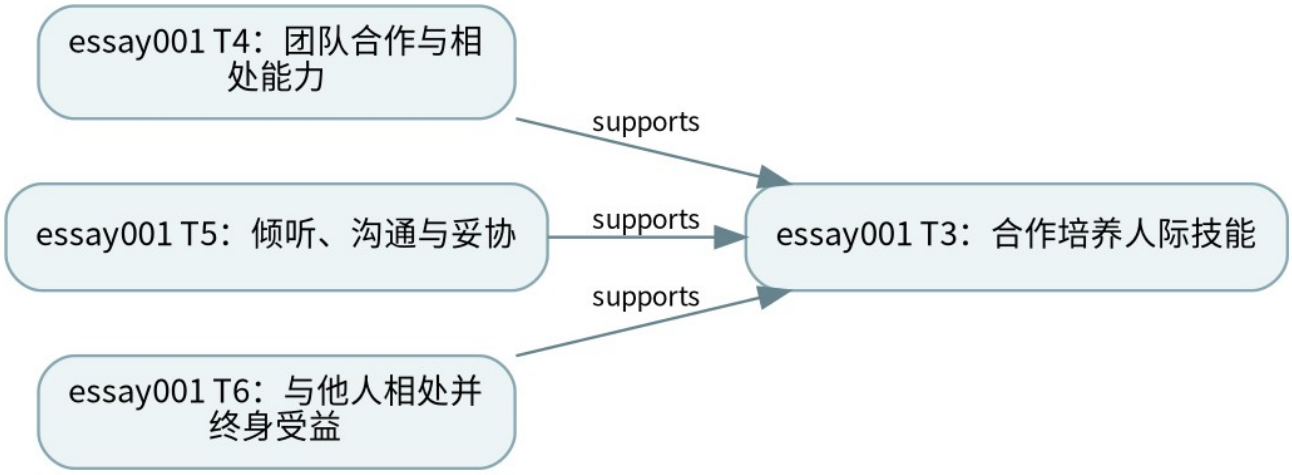
On the official AAEC and CDCP test splits, the complete configuration reaches strict relation F1 scores of 73.29 and 52.02, improving over the inherited configuration by 8.10 and 13.26 points. The aggregate training and inference improvements remain significant after Holm correction over fourteen primary comparisons. Paired-learning estimates are positive on both corpora, but their separate contribution does not pass the family-level correction; joint acceptance and set actions likewise lack confirmed independent advantages. A weak-scoring CDCP condition retains abstention as a failure analysis. Traceable cases from four uploaded corpora illustrate both false-edge removal and true-edge loss, emphasizing that higher fixed utility does not guarantee higher extraction F1.

Keywords: End-to-End Argument Mining; Reliable Correspondence; Paired Learning; Dependency Closure; Graph Repair; Structured Prediction

1.1 □□□□□□□□□□

[1]

AAEC essay001 Premise T4 “What we acquired from team work is not only how to achieve the same goal with others but more importantly, how to get along with others” Claim T3 “through cooperation, children can learn about interpersonal skills which are significant in the future life of all students” T5 T6 T3



1-1 AAEC essay001 I

T3 T4 T4→T3

CPU

1.2 □□□□□□□

AAEC

CDCP

1.2.1

Stab Gurevych [2] Eger [4] Morio [5]

Kawarada [7] Cabessa [8] AASP [9] 2025

GMAM [47] UniASA [48]

DENIM [12] Das [13] FOL-AM [14]

Caputo 2026 [6] ARGUS [10] LLM [11]

1.2.2

[15] CoDa [16] SegMix [17]

QOS [35]

QOS DOS [35] QOS 2026 10 — [49]

ConsistRE [38] CovEReD [37] ODDA
[39] Wang LLM [36]

[40] [18]
“ ”

1.2.3

token [19] AASP [9]

Self-Refine [20] [22] [21]
LLM-refine Claim Evidence Warrant [23]

Re-Cent [43] Pirozelli
[44] Xiao Li ARGUS LLM [24]

ICIC 2026 MCR-BR[34]

F1 [41] F1 [42] [45]
F1

1.2.4

MT-AM [25] AMELIA 19 [26]
[27] “ ”

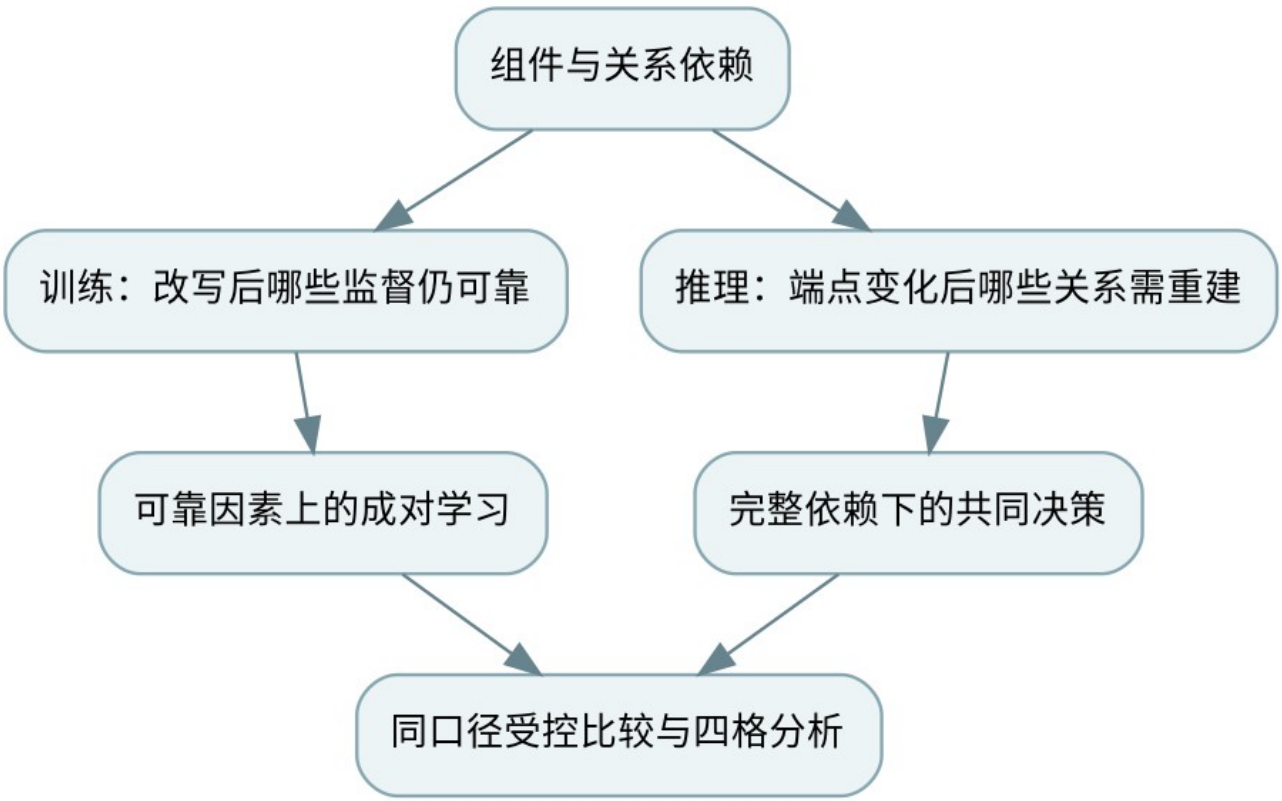
1-1

1-1

QOS[35]

NONE

ConsistRE CovEReD[38][37]



AASP[9]

Re-Cent[43]

MCR-BR[34]

G Kawarada UniASA GMAM DENIM QOS DOS AAEC stance CDCP G A.3

1.3 研究问题与主要研究工作

1-2

1-2

1.3.1

RQ1

RQ2

ICIC

1.3.2

1

2

1-2 “

”

1-2

T1/I0 T0/I0

3-3 5-3

3-4

QOS

3-4

T1/I1 T1/I0

5-3

I1 RelationOnly

4-3 4-4

T1/I1

token

4-4 4-9

T1/I1

4-4 4-6

4-4 4-9

3-6 4-7 4-8a 4-8b

F1 14 2.6.3

1.3.3 MCR-BR

ICIC 2026 MCR-BR [34] 1-3

1-3 MCR-BR

ICIC

MCR-BR

LoRA

C.1

1.4 论文组织结构

2.1 论辩组件与关系表示

2.1.1

$D=(x_1,\ldots,x_n)$ x_i i token x “ D”

$c=(s,e,t)$ $1\le s\le e\le n$ 1 t token token

$r=(c_i,c_j,\ell)$ $G=(C,R)$ “ ”

2.1.2

CDCP $h=(S,c_j,\ell)$ S c_j

1 R [3][46]

2-1

AAEC

MajorClaim Claim Premise

Support Attack

Claim MajorClaim stance

CDCP

Fact Testimony Value Policy Reference

Reason Evidence

Claim/Premise

2-1 [2][3][46] CDCP

E

2.2 合法结构与对象身份

2.2.1

A G_A

AAEC NONE stance CDCP

Reason Evidence NONE

2.2.2

“ ” JSON

A.6 sources

2.3 生成建模与参数高效适配

$Y=(y_1,\ldots,y_L)$

$$P_\theta(Y\mid D)=\prod_{t=1}^{L}P_\theta(y_t\mid D,y_{<t}).$$

Transformer [30] Qwen2.5-7B-Instruct[31] 2.6 A

LoRA [32]

$$W = W_0 + \frac{\alpha}{r} B A , A \in \mathbb{R}^{r \times d_{in}}, B \in \mathbb{R}^{d_{out} \times r}.$$

r α [33] MCR-BR

2.4 候选空间与结构决策

$$G_{rank} = arg\max_{G \in S} E_{rank}(G), G_0 = L_A(G_{rank}).$$

L_A

Candidate Oracle

U V P “ ”

2.5 严格评价与统计口径

TP FP FN P R Precision Recall P R

$$P = \frac{TP}{TP + FP}, R = \frac{TP}{TP + FN}, F_1 = \frac{2TP}{2TP + FP + FN}.$$

micro-F1 micro-F1 CDCP

F1 Relation-F1 Link-F1

FP FN

ID

micro-F1 F1

“±”

2.5.1 ICIC

T0/I0 MCR-BR BR-O

CPU-BIO-SGD T0

2-2

17

29

43

± SD

/pp

AAEC

64.98

64.52

64.81

64.77 ± 0.23

64.83

-0.06

CDCP

38.31

37.88

38.04

38.08 ± 0.22

38.17

-0.09

T0/I0 CDCP AAEC stance

2.5.2

ID F1 TP FP FN T0/I0 T1/I0 T0/I1 T1/I1 “

× × ”

2.6 □□□□□□

2.6.1

AAEC train-test-split.csv ID CDCP train/test AAEC essay171 210 400 essay171 essay209 377 essay209 CDCP 00410 00411

AAEC

train

322

319

4793

3006

0

1

AAEC

test

80

80

1266

809

0

1

CDCP

train

581

580

3901

969

85

8

CDCP

test

150

150

1026

252

34

10

AAEC 11 CDCP 3166 810 24

OOF T0 T1 36 OOF AAEC 255 32 32 CDCP 464 58 58

2.6.2

2-4

Qwen2.5-7B-Instruct

LoRA

rank 16 alpha 32 dropout 0.05

AdamW 2×10^{-4} weight decay 0.01 $\beta=(0.9,0.999)$ $\varepsilon=10^{-8}$

5 epoch 32 4 8

warmup 3% 1 bfloat16

4096 2048 8192 17 29 43

AAEC 500 0.82M token CDCP 900 1.35M token

K=3 1 2 temperature 0.7 top-p 0.9

A100 80GB 64GiB RAM 4 CPU

AAEC 80 CDCP 150 micro-F1

2-5

2 MLP 1024 dropout 0.1

10^{-3} 10 epoch

1 1 0.5 0.5

Platt

wC=wR=1 κC=1 κR AAEC 1 CDCP 0.5 δ=0.20

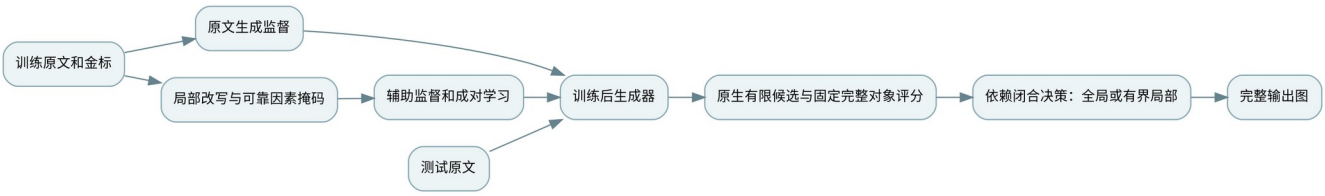
κR∈{0.5,1.0} δ∈{0.01,0.05,0.10,0.20} TP ≤5% G0

24 96 32 3 4

n≤2500 AAEC 3 CDCP 5

50000 4096

Python 3.11 PyTorch 2.5.1 Transformers 4.46.3 PEFT 0.13.2 HiGHS 4



RT-PREREG-01 rt-qwen25-7b-r1 rt-qwen25-tokenizer-r1 epoch NoPair CompPair T1/I0

2.6.3

TP/FP/FN micro-F1 “±” 2000 3 micro-F1

7 F1 ×2 T1/I0−T0/I0 T1/I0−NoPair T1/I1−T1/I0 T1/I1−RelationOnly Single−Sequential T1/I1−Regenerate T1/I1−NoRisk p 14 Holm 95% p

OOF token CPU token A

2.7 训练与推理的接口

2-1 1-2

2-1

ICIC

2.8 本章小结

3.1 引言

NONE

3.2 继承的专项建模基础

3.2.1

Qwen2.5-7B-Instruct LoRA [32]

$W_0 \in \mathbb{R}^{d_{out} \times d_{in}}$

$$h'_t = W_0 h_t + \frac{\alpha}{r} B_g A_g h_t + g_t \frac{\alpha}{r} B_c A_c h_t, A_\cdot \in \mathbb{R}^{r \times d_{in}}, B_\cdot \in \mathbb{R}^{d_{out} \times r}.$$

$g_t \in \{0,1\}$ id type span AAEC CDCP Claim/MajorClaim Policy/Value $g_t = 0$

$r=16\ \ \alpha=32$ rank-32

3.2.2

[33] therefore thus in conclusion consequently hence to conclude

$P_i \in \mathbb{R}^d$

[40] MCR-BR

token EOS

$$L_{gen}(x,G)=-\frac{1}{T_y}\sum_{t=1}^{T_y}\log p_\theta(y_t/x,y_{<t}).$$

ICIC sigmoid $\hat{z}_i$ $z_i=1$

$$L_{sent}=-\frac{1}{N_s}\sum_{i=1}^{N_s}\left[z_i\log\hat{z}_i+(1-z_i)\log(1-\hat{z}_i)\right],L_{ICIC}=L_{gen}+\lambda_{sent}L_{sent}.$$

$\lambda_{sent}=0.5$ L_{ICIC} 3.3 3.4

3.3 □□□□□□□□□□

3.3.1

“ ” “ ”

3-1 AAEC essay001

T3

Claim

[591,714)

through cooperation, children can learn about interpersonal skills which are significant in the future life of all students

T4

Premise

[716,851)

What we acquired from team work is not only how to achieve the same goal with others but more importantly, how to get along with others

T5

Premise

[853,1086)

During the process of cooperation, children can learn about how to listen to opinions of others, how to communicate with others, how to think comprehensively, and even how to compromise with other team members when conflicts occurred

T6

Premise

[1088,1191)

All of these skills help them to get on well with other people and will benefit them for the whole life

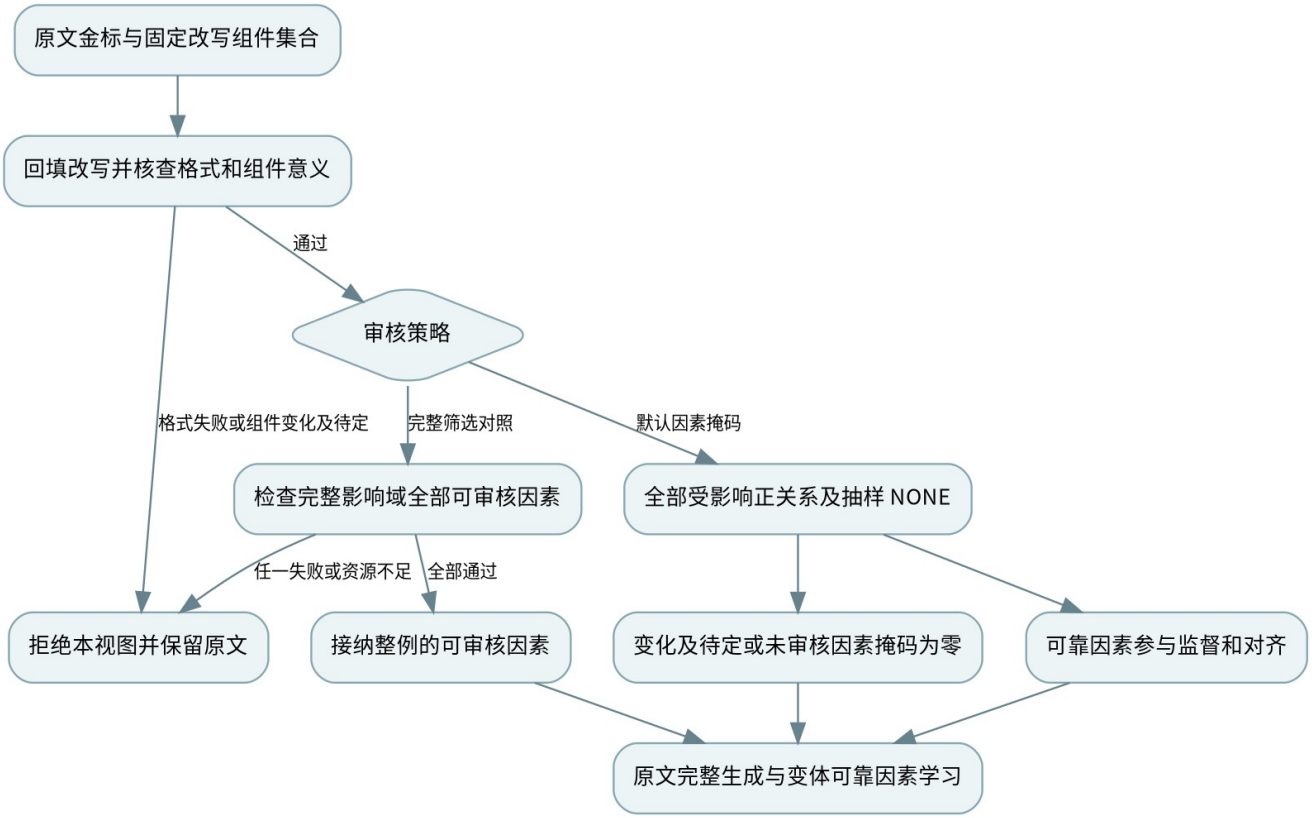
T7

Claim

[1332,1376)

competition makes the society more effective

T4 T5 T6 supports T3 T4→T7 A={T3,T4} T4→T3 T5→T3 T6→T3 T4→T7 NONE



3-1

“ ” “ NONE ”

3.3.4 NONE

$G=(C,R)$ AAEC Claim MajorClaim CDCP Policy Value 17 ID ID center SHA-256 16 17

fallback A

NONE

$\Omega(C)$ $\omega=(S,v)$ S v AAEC $|S)=1$

y_ω NONE 3.3.4

CDCP $\left(\{c_{i_1},\ldots,c_{i_k}\},c_j,\ell\right)$ $\ell\in\{Reason,Evidence\}$

NONE NONE

$V(\omega)=S\cup\{v\}$

$$D_A=\{\omega\in\Omega_{sem}(C):V(\omega)\cap A\neq\varnothing\}.$$

$\Omega_{sem}(C)=\Omega(C)\{U\,¿_{sem}$ 3.3.4 AAEC $\{(u,v):u\in A\vee v\in A,(u,v)\in\Omega_{sem}(C)\}$

A A “ ” NONE

3.3.2

“ NONE”

ID

$$\phi:C\to C',(s_c,e_c,t_c)\mapsto(s'_c,e'_c,t_c).$$

ID ϕ token ϕ $\phi(G)$

3-1 φ

NONE 3.6.4 3.6.5

3.3.3

$J_A\subseteq D_A$ NONE 4 CDCP

NONE

$m_R(\omega)=1$ 0 $m_C(c)$

1 1 “ ”

NONE 1

3.6.5

3.3.4 CDCP NONE

$$h=\left(\{c_1,c_2\},c_3,Reason\right) \quad \left(\{c_1\},c_3,Reason\right)$$

U_{sem} $\Omega_{sem}=\Omega\{U$ ¿$_{sem}$

“ ”

U_{sem} NONE 0 “ ”

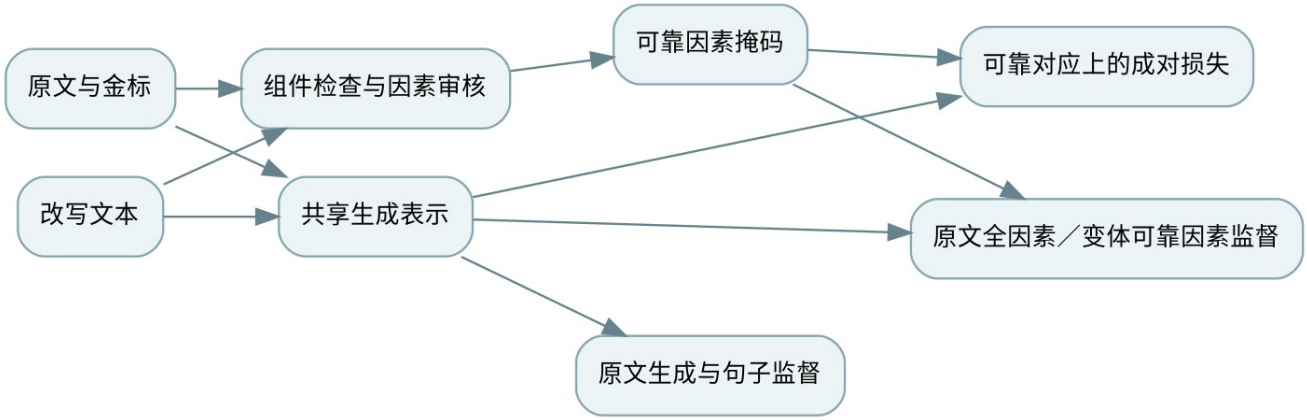
0 NONE

3.4 □□□□□□□□□□

“ ” 3-1 c_2

c_1 c_3

3-2 ϕ



3-2

3-2 3.4.3

3.4.1

h_i^x $B_x(c)$ h_{end}^x F.2

$$u_c^x=MeanPool_{i\in B_x(c)}h_i^x,\ p_C^x(c)=softmax\,f_C\left(\left[u_c^x;h_{end}^x\right)\right).$$

token h_{end}^x

$$p_R^x(S,v)=softmax\,f_R\left(\left[Pool\left(\{u_c^x:c\in S\}\right);u_v^x;h_{end}^x;\eta(S,v)\right)\right),$$

f_C f_R $Pool$ η token

NONE

3.4.2

$$A^{+} = A \cup \bigcup_{\substack{\omega \in D_A \\ y_\omega \neq NONE}} V(\omega).$$

NONE x' A^{+} $\{c : m_C(c) = 1\}$

$$L_C^{align}(x, x') = \frac{1}{|A^{+}|} \sum_{c \in A^{+}} JS\left(p_C^{x}(c), p_C^{x'}(\phi(c))\right),$$

$JS(p, q) = 1/2\, KL(p \,//\, \acute{p}) + 1/2\, KL(q \,//\, \acute{p})$ $\acute{p} = (p+q)/2$

$\{\omega \in J_A : m_R(\omega) = 1\}$ D_+ NONE D_0 $B = \{b \in \{+, 0\} : |D_b| > 0\}$

$$L_R^{align}(x, x') = \frac{1}{|B|} \sum_{b \in B} \frac{1}{|D_b|} \sum_{\omega \in D_b} JS\left(p_R^{x}(\omega), p_R^{x'}(\phi(\omega))\right).$$

NONE A^{+}

JS x M_C M_R M_R^{+} M_R^{0} NONE

$B_M = \{b \in \{+, 0\} : |M_R^b| > 0\}$

$$S(x; M_C, M_R) = \frac{1}{|M_C|} \sum_{c \in M_C} CE\left(p_C^{x}(c), t_c\right) + \frac{1}{|B_M|} \sum_{b \in B_M} \frac{1}{|M_R^b|} \sum_{\omega \in M_R^b} CE\left(p_R^{x}(\omega), y_\omega\right),$$

CE t_c $M_C = \varnothing$ 0 $B_M = \varnothing$ 0 $S(x, G) = S(x; C, \Omega_{sem}(C))$

$S_m(x', \phi(G)) = S(x'; \{\phi(c) : m_C(c) = 1\}, \{\phi(\omega) : \omega \in J_A, m_R(\omega) = 1\})$ 1

NONE

3.4.3

d m_d

$$L_{heads,d}^{sup} = \frac{S(x_d, G_d) + \sum_{j=1}^{m_d} S_{m_j}\left(x'_{dj}, \phi_j(G_d)\right)}{1 + m_d}.$$

$$L_d = L_{ICIC}(x_d, G_d) + \lambda_{sup} L_{heads,d}^{sup} + \lambda_C \acute{L}_{C,d}^{align} + \lambda_R \acute{L}_{R,d}^{align}.$$

$\lambda_{sup} = 1$ $\lambda_C = \lambda_R = 0.5$ m_C 1 m_d

$1 + m_d$ $1/(1 + m_d)$ L_{ICIC}

LoRA

3.5 □□□□□

3.5.1

3-1 2.6

□□ 3-1 □□□□□□□□□□□□□□

```
                D_train
              B_try        B_accept            U
        mode ∈ {FactorMask, FullFilter}
                  θ
01
02  for          d ∈ D_train do
03          x_d            G_d     Views_d = ∅
04     3.3.1
05       Ω(C_d)                   Ω_sem             A
06              D_A     attempts = 0
07   while attempts < B_try    |Views_d| < B_accept do
08    attempts ← attempts + 1
09                                φ
10    if                        then
11
12     continue
13    end if
14
15    if D_A ≠ ∅ then
16     if mode = FullFilter then J_A ← D_A
16a     else J_A ←              ∪              NONE
16b       J_A                                   m_R
17    end if
18    if                                      then
19
20     continue
21    end if
22              φ(G_d)  φ                  Views_d
23   end while
24       Views_d
25  end for
26  for step = 1, ..., U do
27
28
29
30     φ
31
32     3.4.3
33   if     2.6                then
34    break
35   end if
```

36 end for

37

LoRA

3.5.2

性质一：标注输运保持。 ϕ ϕ G $\phi(G)$

性质二：单因素预测稳定的充分条件。 y p $\delta = p_y - max_{k \neq y} p_k > 0$ q $\| p - q \|_\infty < \delta / 2$ $arg\,max\, q = y$ $k \neq y$ $q_y - q_k > p_y - p_k - \delta \geq 0$ JS

3.6 实验结果与分析

3.6.1

T0/I0 Heads Plain QOS NoPair CompPair T1/I0 FullFilter RandomCenter I0 500 900

NoPair CompPair T1/I0 A NONE CDCP NONE

3-2

T0I0

—

Heads

—

Plain QOS

NoPair

CompPair

T1I0

FullFilter RandomCenter

3.6.2

3-3 I0

AAEC	F1	
AAEC	F1	
CDCP	F1	
CDCP	F1	
T0/I0		
84.18 ± 0.79		
65.18 ± 3.43		
75.99 ± 1.61		
38.76 ± 3.37		
84.36 ± 0.82		
65.60 ± 3.21		
76.25 ± 1.58		
39.02 ± 3.37		
84.76 ± 1.17		
65.64 ± 2.59		
76.93 ± 1.45		
39.15 ± 2.39		

QOS

85.04 ± 1.08

66.21 ± 3.37

77.49 ± 1.92

40.08 ± 2.78

85.49 ± 0.88

66.83 ± 3.03

77.97 ± 2.01

41.14 ± 2.70

86.15 ± 0.78

67.82 ± 2.60

78.65 ± 1.51

42.06 ± 2.86

T1/I0

86.73 ± 0.67

68.81 ± 2.05

79.92 ± 1.51

43.65 ± 3.53

86.49 ± 0.88

68.36 ± 2.35

79.56 ± 1.76

43.92 ± 2.82

86.26 ± 0.97

67.99 ± 2.66

79.60 ± 1.32

44.05 ± 3.39

T1/I0 T0/I0 AAEC 3.63 pp CDCP 4.89 pp Holm p 0.0120 “±”

3.6.3

3-4 F1

/pp

95% CI

17 / 29 / 43

p

AAEC

T1I0 − T0I0

3.63

[1.78, 5.60]

1.85 / 4.45 / 4.57

0.0120

Holm

AAEC

T1I0 – NoPair

1.98

[0.69, 3.43]

0.74 / 2.47 / 2.72

0.0600

Holm

AAEC

CompPair – NoPair

0.99

[0.24, 1.95]

0.37 / 1.36 / 1.24

0.0295

p

AAEC

T1I0 – CompPair

0.99

[0.20, 1.97]

0.37 / 1.11 / 1.48

0.0325

p

AAEC

NoPair – QOS

0.62

[0.00, 1.80]

0.12 / 0.99 / 0.74

0.1354

p

AAEC

NoPair – Plain

1.19

[0.22, 2.51]

1.48 / 1.61 / 0.49

0.0405

p

AAEC

FullFilter − T1I0

-0.45

[-1.31, 0.08]

0.00 / -0.74 / -0.62

0.2039

p

AAEC

RandomCenter − T1I0

-0.82

[-1.84, 0.00]

-0.12 / -0.99 / -1.36

0.0775

p

CDCP

T1I0 − T0I0

4.89

[2.91, 7.23]

5.95 / 4.37 / 4.37

0.0120

Holm

CDCP

T1I0 − NoPair

2.51

[0.98, 4.39]

2.78 / 3.17 / 1.59

0.0720

Holm

CDCP

CompPair − NoPair

0.93

[0.13, 1.94]

0.79 / 1.19 / 0.79

0.0585

p

CDCP

T1I0 – CompPair

1.59

[0.14, 3.21]

1.98 / 1.98 / 0.79

0.0495

p

CDCP

NoPair – QOS

1.06

[0.00, 2.85]

1.98 / 0.40 / 0.79

0.1119

p

CDCP

NoPair – Plain

1.98

[0.60, 4.14]

3.17 / 1.59 / 1.19

0.0360

p

CDCP

FullFilter – T1I0

0.26

[-1.10, 1.59]

-0.40 / 0.00 / 1.19

0.6747

p

CDCP

RandomCenter – T1I0

0.40

[-0.61, 1.60]

-0.40 / 0.79 / 0.79

0.5032

p

NoPair 95% Holm p 0.0600 0.0720

QOS

3.6.4

A B stable changed undecided original_disagreement C 60 κ 0.86 300 300 πh=nh/Nh

3-5

Wilson 95%

κ

AAEC

320

300

4

1.33%

[0.52, 3.38]

94.69%

0.690

CDCP

320

300

3

1.00%

[0.34, 2.90]

95.94%

0.755

Nh

n

π

Wilson 95%

κ

AAEC

683

300

0.4392

280

3

1.07%

[0.37, 3.10]

96.00%

0.771

AAEC

153

153

1.0000

123

2

1.63%

[0.45, 5.74]

92.16%

0.804

AAEC NONE

877

300

0.3421

279

4

1.43%

[0.56, 3.63]

90.67%

0.591

AAEC NONE

2000

300

0.1500

266

1

0.38%

[0.07, 2.10]

94.67%

0.785

CDCP

315

300

0.9524

282

2

0.71%

[0.19, 2.55]

93.00%

0.606

CDCP

290

290

1.0000

249

4

1.61%

[0.63, 4.06]

94.14%

0.811

CDCP NONE

470

300

0.6383

277

5

1.81%

[0.77, 4.16]

93.00%

0.697

CDCP NONE

3251

300

0.0923

269

2

0.74%

[0.20, 2.67]

94.33%

0.776

κ NONE κ 5%

GPU

token/M

GPU

GiB

AAEC

T0I0

1595

50

0

0.00 0.00

0

1.69

47.29

AAEC

Heads

1595

50

0

0.00 0.00

0

1.93

51.52

AAEC

NoPair

1595

50

500

1.14 0.82

1210

3.11

62.37

AAEC

CompPair

1595

50

500

1.14 0.82

1210

3.26

65.54

AAEC

T1I0

1595

50

500

1.14 0.82

1210

3.53

64.85

CDCP

T0I0

2900

95

0

0.00 0.00

0

2.86

47.57

CDCP

Heads

2900

95

0

0.00 0.00

0

3.41

50.80

CDCP

NoPair

2900

95

900

2.02 1.35

2190

5.33

62.42

CDCP

CompPair

2900

95

900

2.02 1.35

2190

5.69

64.95

CDCP

T1I0

2900

95

900

2.02 1.35

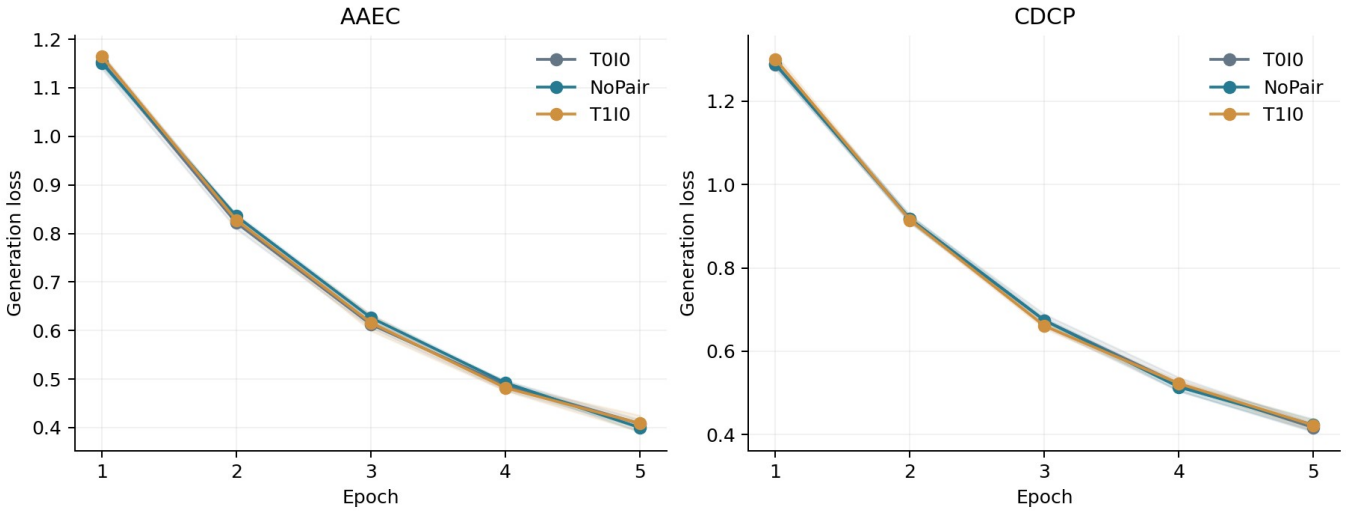
2190

5.98

65.61

5N 5×ceil(N/32) 44,310,528 63,460,376 54 5 epoch 270 epoch

JS JS LoRA NoPair CompPair T1 3-3 B



3-3

3.6.5

3-8

95%

95%

AAEC

1053 257

1.05%

0.81%

[0.31, 1.45]

5.83%

[4.19, 7.60]

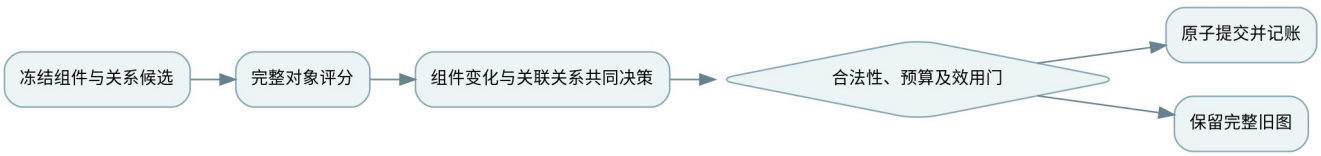
60 998

CDCP

1190 401

1.21%

0.91%



[0.26, 1.79]

5.98%

[4.06, 8.10]

68 1132

“ ” “ ” undecided original_disagreement 2000 factor_population

H

3-3

3.6.6

3-1 essay001 T4 T5 T6 T3

CDCP 00662 4.6.7 value policy reason “ ”

3.7 本章小结

F1 Holm

4.1 引言

MCR-BR [34] I1 BR-J

AAEC essay098 Premise [1114,1267) supports [1020,1099) F1 4.6.5—4.6.6

4-1

[46] [43][44]

4.2 候选生成与初始图

MCR

4.2.1

x K $K=3$ temperature 0 temperature 0.7 top-p 0.9 E ICIC $K=3$ K

$$C=\{C^{(1)},\dots,C^{(K)}\}.$$

$C^{(k)}$ $G^{(k)}=(C^{(k)},R^{(k)})$

4.2.2

MCR-BR MCR

$$E_{rank}(G) = 0.4\,E_{dens}(G) + 0.4\,E_{coh}(G) + 0.2\,E_{val}(G).$$

$Q(G)$ $p_r \in \{0,1\}$ $d_{coh}^{in}(j)$ j

$$E_{dens}(G) = \frac{1}{|Q(G)|} \sum_{j \in Q(G)} min\left\{ \frac{d_{coh}^{in}(j)}{\tau_j}, 1 \right\}, E_{coh}(G) = \frac{1}{|R|} \sum_{r \in R} p_r.$$

AAEC Claim CDCP Policy/Value MajorClaim

$\tau_j = 2$ $\tau_j = 1$ CDCP MCR-BR

E_{val} 1 0

G_0 ID ID

p_r

4.2.3

U

OOF

A.6

2 $T_*(u)$ $V = \{(u,t) : t \in T_*(u)\}$ NULL

ID A.6

4.2.4

P AAEC CDCP 64 2 3

256

N_{slot} NONE

CasiMedicos-Arg CDCP CDCP

A.3 D.3

4.2.5

4096 64 2 CDCP 256

50000

m q AAEC $m(m-1)q^2$ CDCP $\Sigma_{k=1}^{3} C(m,k)(m-k)q^{k+1}$

CasiMedicos C D CDCP

4.3 组件集合替换与受影响关系重建

4.3.1

C B D_0 D_0 D x $B \cap C$ B D_0

$$D = D_0 \cup \{ c \in C : \exists b \in B, Conflict(c, b) \}, C' = (C ¿) \cup B.$$

B D

$0 \to 1$ $1 \to 0$ $1 \to 1$ $1 \to 2$ $2 \to 1$ $|D| + |B| \le 3$

4.3.2

“ ” “ ”

NONE NONE

$A = D \cup B$ P_A P A

NONE

S_R

$$P_{tx} = P_A \cup S_R.$$

RelationOnly $D = B = \varnothing$ S_R S_R 4.2.4

C' $G' = (C', R')$ C'

S_R

CDCP “ ” “ ” a, b, c, t $h = (\{a, c\}, t, Reason)$

b h b h a, c, t h P_A S_R h

h h S_R h b h

essay098

4.4 对象评分与事务接受

4.4.1

$$q_C(c/x), q_R(r/x).$$

NULL/

MLP MLP

2.4

q

4.6.4

B

4.4.2

out-of-fold OOF

OOF

NONE

2.6.1

OOF 2.6 OOF OOF

4.6.4 4.6.5

4.4.2.1

F1 TP δ {0.01,0.05,0.10,0.20} TP

5% κR=1 δ κR=0.5 κ

TP F1 δ κ

OOF

4.4.3

$X \in \{C, R\}$ $0 < \kappa_X < 2$

$$\Phi_X(G) = 2 \sum_{o \in X(G)} q_X(o/x) - \kappa_X \left| X(G) \right|, \Phi(G) = w_C \Phi_C(G) + w_R \Phi_R(G),$$

$w_C, w_R > 0$ $w_C = w_R = 1$ κ_X F1

Y_X $q_X(o/x) = P(o \in Y_X / x)$ $FP_X = \left| X(G) \right) - TP_X$ $FN_X = \left| Y_X \right) - TP_X$

$$\Phi_X(G) = (2 - \kappa_X) E\left[\left| Y_X \right) / x \right) - E\left[\kappa_X FP_X + (2 - \kappa_X) FN_X / x \right).$$

F1 F1 [41] F1 F1

[42]

4.4.4

$$\Delta\Phi_C \geq 0, \Delta\Phi_R \geq 0, \Delta\Phi > \delta > 0.$$

F1 δ

$d_C = |D) + |B)$ $d_R = |R \triangle R')$ “ ”

G_0

4.4.5

AAEC essay098 Premise[1114,1267) Premise[1020,1099) supports

4.6.5 35 4.6.6

4.4.4 B.6

4.5 □□□□□□□

4.5.6 4.5.7

4.5.1

“ ILP ” C' $z_c = I[c \in C']$ $y_r \in \{0,1\}$ $r \in P$

$$\max_y w_C \Phi_C(C') + w_R \sum_{r \in P} (2 q_R(r/x) - \kappa_R) y_r.$$

$$\sum_{t \in T(u)} z_{(u,t)} \leq 1, \quad ¿ z_c + z_d \leq 1 \quad Conflict(c,d);$$
$$y_r \leq z_c \; \forall c \in Ends(r), \quad ¿ \sum_{r \in \Gamma_s} y_r \leq 1 \; \forall s;$$
$$y_r = I[r \in R) \; \forall r \notin P_{tx}, \quad ¿ G' \in G_A.$$

Γ_s NONE G_A

$0 \leq o_c \leq M - 1$ $M = |V)$

$$o_v \geq o_u + 1 - M(1 - y_{(u,v,\ell)}).$$

CDCP C'

$y_h = 0$

ILP $y_r^0 = I[r \in R)$

$$d_R = \sum_{r: y_r^0 = 1} (1 - y_r) + \sum_{r: y_r^0 = 0} y_r.$$

d_C, d_R $\Delta\Phi_C, \Delta\Phi_R \geq 0$ $\Delta\Phi > \delta$

incumbent B

4.5.2

S_R

$(|D), |B))$ $2 \to 1$

B.4

32 3 4 128 12 24
96 BR-J

B.3 B.4.3

4-1

MCR
if then
return
end if
G ← G0
for do

while do
Q Gnoop ← G H ← ∅
for Q do
1 4-2
H
end for
if H then

Gnoop
Gnoop
1
else
Gnoop
end if
end while
end for
return G

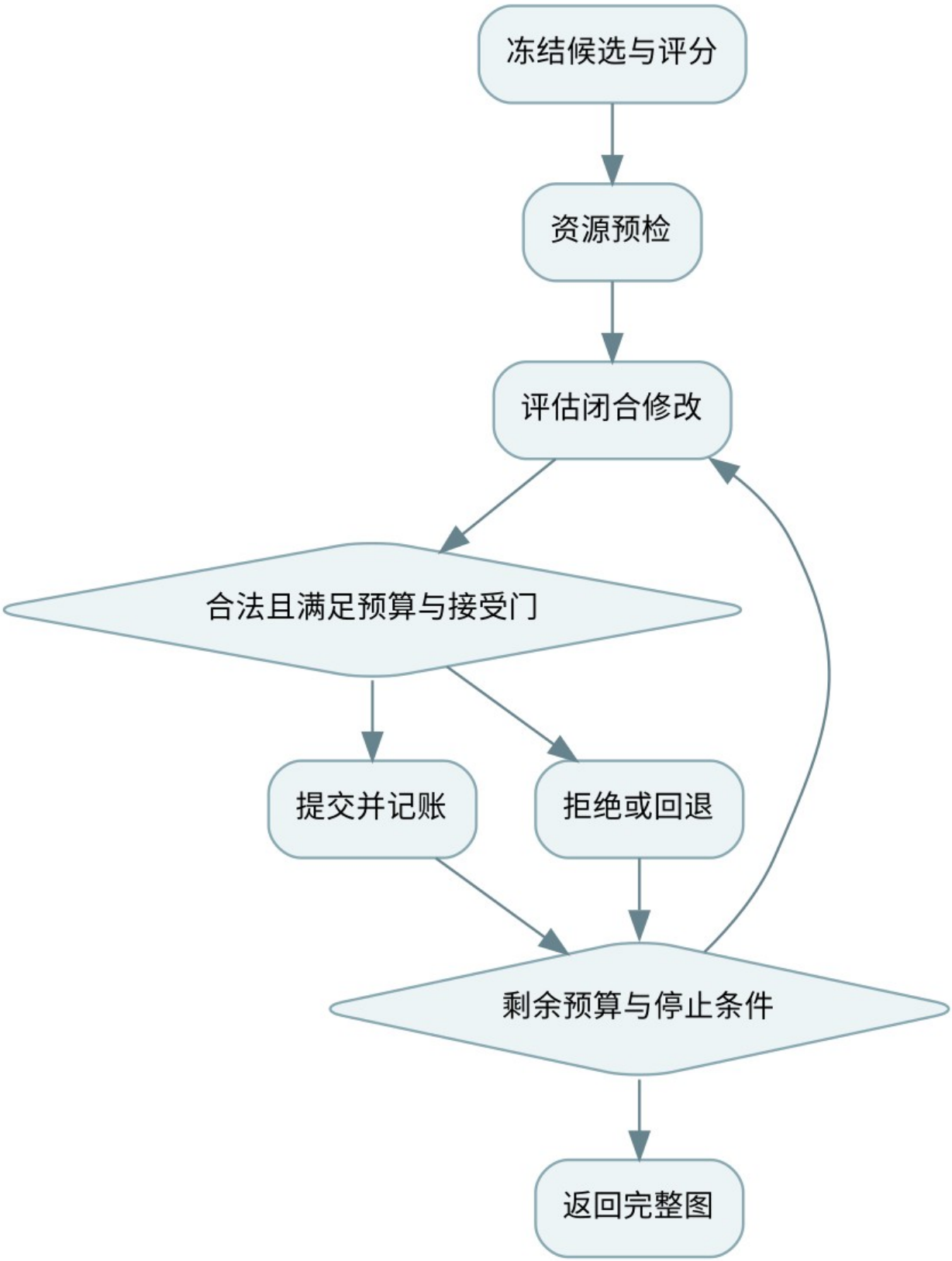
4-2

G

G
if then
return
end if

ILP
incumbent
if then
return
end if

return



4-2 BR-J

4-2 4-1 4-2 no-op ILP

incumbent B

4.5.3

4-1

$X \in \{C, R\}$ $a_X(o) = 2q_X(o/x) - \kappa_X$

$$\Delta\Phi_X = \sum_{o \in X(G') \setminus (G)} a_X(o) - \sum_{o \in X(G) \setminus (G')} a_X(o).$$

B.6

4.5.4

4-2

B.6 F1

F1

4.5.5

4-3 Φ

B.6

micro-F1 T P_0 Q $F = 2T/(P_0 + Q)$

$$F' - F = \frac{2\Delta T - F\Delta P_0}{P_0 + Q + \Delta P_0}.$$

ΔT F κ

F1 F1

4.5.6

4.5.1 C' CDCP

G_0

1 $v \in V$ $z_v \in \{0,1\}$ $r \in P$ $y_r \in \{0,1\}$ z_v^0 y_r^0 G_0

$a_X(o) = 2q_X(o/x) - \kappa_X$

$$\max_{z,y} \Phi(z, y) = w_C \sum_{v \in V} a_C(v) z_v + w_R \sum_{r \in P} a_R(r) y_r.$$

2

$$\sum_{t \in T_*(u)} z_{(u,t)} \le 1 \; \forall u \in U, \qquad z_c + z_d \le 1 \quad Conflict(c, d);$$

$$y_r \le z_c \; \forall r \in P, \forall c \in Ends(r), \qquad \sum_{r \in \Gamma_s} y_r \le 1 \; \forall s;$$

$$y_h + z_w \le 1 \; \forall h \in P_{CDCP}, \forall w \in Btw(h).$$

CDCP $h = (S, v, \ell)$ $Btw(h)$

S S “S ” $Btw(h)$

4.5.1

4.3.2 a, b, c, t $y_h + z_b \le 1$ b $h = (\{a, c\}, t, Reason)$ h b

3 G_0

$$d_C^{net} = \sum_{v:z_v^0=1} (1 - z_v) + \sum_{v:z_v^0=0} z_v \le B_C, d_R^{net} = \sum_{r:y_r^0=1} (1 - y_r) + \sum_{r:y_r^0=0} y_r \le B_R.$$

$|D)+|B)$ d_R^{net} G_0 $\Phi_C(z) \ge \Phi_C(z^0)$

$\Phi_R(y) \ge \Phi_R(y^0)$ $B_C = 24$ $B_R = 96$ $\Phi^* - \Phi(z^0, y^0) > \delta$ G_0

(z^0, y^0) G_0 “ ” “

”

4 G_0

4-1

4-1

BR-J

C' y_r $r \in P_{tx}$

z_v y_r

Φ

Φ

CDCP

C' $y_h = 0$

$y_h + z_w \le 1$ $w \in Btw(h)$

G_0

G_0

24 96

24 96

ILP 128

G_0

5 Φ $\Phi \geq \Phi^* - \varepsilon$ $d_C^{net} + d_R^{net}$ G_0 ε

“ ”

4.5.7

“ ” $V \cup P$ G_0 4.4.1 “

”

$n = |V| + |P|$ $\sum_h |Btw(h)|$ $n \leq 2500$

2500 AAEC CDCP P95 2.71 4.12 3 5 3000 H

AAEC 3 CDCP 5 δ G_0 incumbent

$\Phi - \Phi(G_0) > \delta$ “ ” gap G_0 incumbent “ ” G_0 “ ”

G_0

4-1 4-2

F1

I1 “ BR-J”

4.6 □□□□□

4.6.1

T1 RelationOnly Sequential Single SetLocal Global I1 2500 I1 Global

Regenerate token NoRisk δ NoCalibration MissingLow

4-2

RelationOnly

Sequential Single

SetLocal Global

NoRisk

δ

NoCalibration

MissingLow

0

Regenerate

4.6.2

4-3 T1

AAEC F1

AAEC F1

CDCP F1

CDCP F1

T1/I0

86.73 ± 0.67

68.81 ± 2.05

79.92 ± 1.51

43.65 ± 3.53

RelationOnly

86.73 ± 0.67

72.61 ± 2.11

79.92 ± 1.51

50.67 ± 2.65

87.31 ± 0.54

72.93 ± 2.15

81.74 ± 1.54

51.94 ± 3.13

87.30 ± 0.51

73.12 ± 1.80

81.87 ± 1.31

51.94 ± 2.54

87.30 ± 0.51

73.12 ± 1.80

81.87 ± 1.31

51.94 ± 2.54

87.60 ± 0.44

73.29 ± 1.76

81.91 ± 1.22

52.02 ± 2.59

T1/I1

87.60 ± 0.44

73.29 ± 1.76

81.91 ± 1.22

52.02 ± 2.59

86.84 ± 2.23

69.22 ± 2.24

81.06 ± 0.59

44.18 ± 2.19

87.63 ± 0.41

73.29 ± 1.76

81.95 ± 1.23

52.02 ± 2.59

88.02 ± 0.36

73.73 ± 1.87

81.94 ± 1.24

52.05 ± 2.64

87.60 ± 0.44

73.29 ± 1.76

81.91 ± 1.22

52.02 ± 2.59

/pp

95% CI

17 / 29 / 43

p

AAEC

T1I1 – T1I0

4.48

[3.43, 5.54]

3.88 / 4.90 / 4.66

0.0070

Holm

AAEC

T1I1 – RelationOnly

0.67

[0.23, 1.37]

0.53 / 0.40 / 1.09

0.1924

Holm

AAEC

Single – Sequential

0.18

[-0.33, 0.92]

-0.01 / -0.03 / 0.59

1.0000

Holm

AAEC

T1I1 – Regenerate

4.07

[-1.65, 9.40]

7.71 / 2.67 / 1.81

0.6797

Holm

AAEC

T1I1 – NoRisk

0.00

[0.00, 0.00]

0.00 / 0.00 / 0.00

1.0000

Holm

AAEC

SetLocal – Single

0.00

[0.00, 0.00]

0.00 / 0.00 / 0.00

1.0000

p

AAEC

Global – SetLocal

0.17

[-0.06, 0.45]

0.27 / 0.01 / 0.23

0.1294

p

AAEC

Global – Single

0.17

[-0.06, 0.45]

0.27 / 0.01 / 0.23

0.1294

p

AAEC

T1I1 – MissingLow

0.00

[0.00, 0.00]

0.00 / 0.00 / 0.00

1.0000

p

AAEC

T1I1 – NoCalibration

-0.45

[-0.88, -0.09]

-0.53 / -0.48 / -0.32

0.0300

p

CDCP

T1I1 – T1I0

8.37

[6.11, 10.94]

7.14 / 8.20 / 9.77

0.0070

Holm

CDCP

T1I1 – RelationOnly

1.35

[0.45, 2.49]

1.54 / 1.20 / 1.31

0.1239

Holm

CDCP

Single – Sequential

0.00

[-0.92, 1.17]

0.08 / -0.63 / 0.56

1.0000

Holm

CDCP

T1I1 – Regenerate

7.84

[-0.75, 15.48]

9.92 / 10.58 / 3.02

0.3508

Holm

CDCP

T1I1 – NoRisk

0.00

[0.00, 0.00]

0.00 / 0.00 / 0.00

1.0000

Holm

CDCP

SetLocal – Single

0.00

[0.00, 0.00]

0.00 / 0.00 / 0.00

1.0000

p

CDCP

Global – SetLocal

0.08

[-0.50, 0.50]

-0.33 / 0.35 / 0.22

0.7151

p

CDCP

Global – Single

0.08

[-0.50, 0.50]

-0.33 / 0.35 / 0.22

0.7151

p

CDCP

T1I1 – MissingLow

0.00

[0.00, 0.00]

0.00 / 0.00 / 0.00

1.0000

p

CDCP

T1I1 – NoCalibration

-0.03

[-0.48, 0.22]

-0.21 / 0.00 / 0.11

0.9115

p

I1 I0 AAEC CDCP 4.48 8.37 pp Holm RelationOnly
Single–Sequential F1

SetLocal Single Global F1 690
gap 0 B

I1 I1 F1 F1 AAEC TP
3 5

4.6.3

4-5 — —

TP

AAEC

3798

3781

2427

2411

2391

1788

1704

1704

CDCP

3078

3031

756

734

705

399

358

358

× × 3798 3078 2427 756 “ ” “ ”
TP

candidate_complete CDCP 5.4

4.6.4

4-6

TP

TP

TP

FP

FP

TP

F1

AAEC

T1I1

1670

37

3

244

6

99.82%

4 240

AAEC

RelationOnly

1670

26

3

220

6

99.82%

4 240

AAEC

Single

1670

37

3

233

6

99.82%

4 240

AAEC

Sequential

1670

28

3

237

6

99.82%

4 240

AAEC

NoRisk

1670

37

3

244

6

99.82%

4 240

AAEC

NoCalibration

1670

39

5

272

6

99.70%

5 240

AAEC

Regenerate

1670

516

506

680

670

69.70%

99 240

CDCP

T1I1

330

29

1

170

6

99.70%

2 450

CDCP

RelationOnly

330

21

1

157

6

99.70%

3 450

CDCP

Single

330

30

1

165

6

99.70%

2 450

CDCP

Sequential

330

25

2

182

6

99.39%

1 450

CDCP

NoRisk

330

29

1

170

6

99.70%

2 450

CDCP

NoCalibration

330

30

1

168

6

99.70%

2 450

CDCP

Regenerate

330

178

174

345

341

47.27%

88 450

AAEC CDCP 4/240 2/450 — 5% 4-6 4-8b

4-7

AUROC

AP

Brier

Brier

ECE10

n

AAEC

calibration

C

3253

1403

0.8967

0.8490

0.0909

0.2453

0.0083

1412

0.880

AAEC

calibration

R

14118

849

0.8641

0.5995

0.0263

0.0566

0.0036

855

0.747

AAEC

strategy

C

3041

1321

0.8909

0.8453

0.0962

0.2457

0.0175

1333

0.872

AAEC

strategy

R

12532

803

0.8659

0.6082

0.0285

0.0600

0.0032

805

0.742

CDCP

calibration

C

3607

1041

0.8660

0.7323

0.0985

0.2053

0.0084

1061

0.795

CDCP

calibration

R

10972

266

0.7403

0.3391

0.0176

0.0237

0.0010

271

0.506

CDCP

strategy

C

3837

1092

0.8748

0.7454

0.0933

0.2036

0.0150

1101

0.809

CDCP

strategy

R

13326

226

0.8013

0.3649

0.0118

0.0167

0.0068

239

0.548

4-7 AUROC AP Brier ECE

4-8a

n

AUROC

AP

Brier Brier

ECE10

n

q

AAEC/C

8779 3781

0.898

0.853

0.0913 0.2452

0.0081

0.5

3810 3349

0.879 0.879

AAEC/R

41876 2391

0.863

0.602

0.0250 0.0538

0.0033

0.5

2378 1788

0.752 0.736

CDCP/C

10525 3031

0.875

0.761

0.0892 0.2050

0.0146

0.5

3091 2525

0.817 0.793

CDCP/R

41818 705

0.789

0.368

0.0116 0.0166

0.0059

0.25

714 399

0.559 0.512

AAEC

Sequential

160

22 22

34 240

28

82.35%

3

1.25%

AAEC

Single

160

16 31

43 236

37

86.05%

3

1.27%

AAEC

SetLocal

160

16 31

43 236

37

86.05%

3

1.27%

AAEC

T1I1

143

26 49

43 247

37

86.05%

3

1.21%

CDCP

Sequential

182

68 68

31 184

25

80.65%

2

1.09%

CDCP

Single

187

52 81

36 166

30

83.33%

1

0.60%

CDCP

SetLocal

187

52 81

36 166

30

83.33%

1

0.60%

CDCP

T1I1

159

51 86

35 171

29

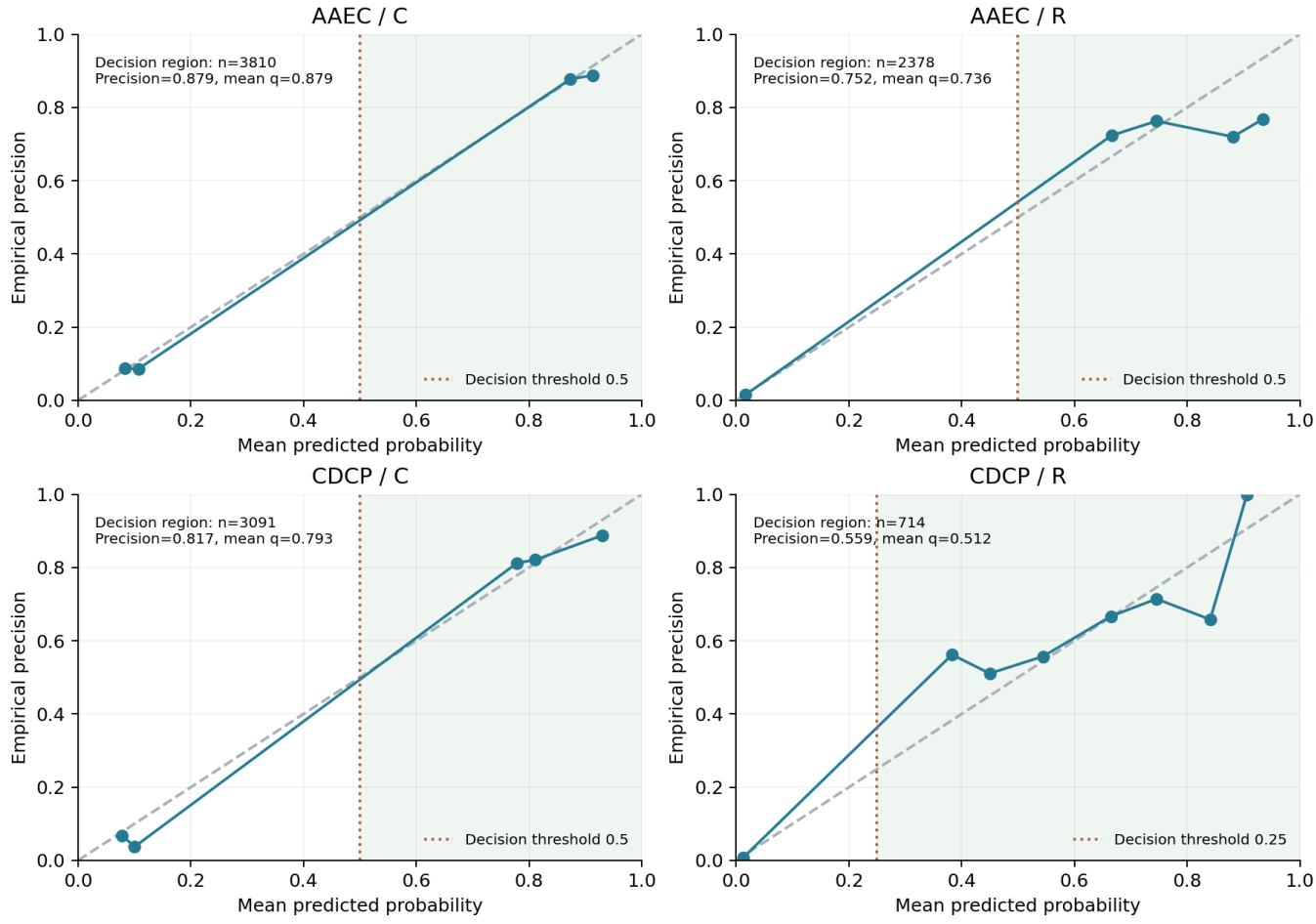
82.86%

1

0.58%

AAEC 0.5 CDCP 0.25 0.5 q ECE

AAEC 37/43 CDCP 29/35



0	MissingLow	I1	60	NaN

P95

P95

AAEC

T0I0

17.58

0.00

0.00

0.70

0.90

18.28

21.47

0/240

AAEC

T1I0

17.58

0.00

0.00

0.68

0.90

18.26

21.40

0/240

AAEC

Single

17.58

0.93

12.93

0.47

0.61

31.91

37.60

0/240

AAEC

SetLocal

17.58

0.93

12.93

1.28

1.70

32.72

38.25

0/240

AAEC

T1I1

17.58

0.93

12.93

0.73

0.96

32.17

37.78

0/240

AAEC

Regenerate

31.44

0.00

0.00

0.05

0.07

31.49

37.21

0/240

CDCP

T0I0

24.45

0.00

0.00

1.12

1.46

25.57

30.02

0/450

CDCP

T1I0

24.45

0.00

0.00

1.11

1.46

25.56

30.02

0/450

CDCP

Single

24.45

1.35

17.92

0.75

0.97

44.46

52.55

0/450

CDCP

SetLocal

24.45

1.35

17.92

2.10

2.74

45.82

53.75

0/450

CDCP

T1I1

24.45

1.35

17.92

1.18

1.53

44.89

52.93

0/450

CDCP

Regenerate

43.71

0.00

0.00

0.08

0.11

43.79

51.88

0/450

I1	Regenerate	AAEC	4096 I0	2048	token	CDCP	6144 H	3072	2

4.6.5 AAEC

essay098 f68e231bdf7576fe87d57e801350c2b0ef2f513d51e29938b223671a392f528c CPU-BIO-SGD
17 1 54 — AAEC

4-10
AAEC
essay098

TP/FP/FN

TP/FP/FN

F1

F1

FP

TP

17

1 / 54 / 5

1 / 19 / 5

3.28

7.69

35

0

29

1 / 45 / 5

0 / 17 / 6

3.85

0.00

28

1

4-11 essay098
supports

1

Premise

[1114,1267)

if a child works as a shop assistant, he will have to keep an eye on good and products, calculate prices correctly and take responsibilities for any lost

Premise

[1020,1099)

To keep their jobs, children will have to try their best to finish their duties

17 4-11 35 F1 3.28 7.69 supports Premise[432,469)“children can learn about the hardship” Claim[313,320)“measure” Claim[526,530)“such”

Qwen CPU

4.6.6

29 essay098 4-11 TP/FP/FN 1/45/5 0/17/6 F1 0 28

690

4.6.7 CDCP

4-12 CDCP 00662 reason

1

value

[0,53)

I generally prefer e-mail to telephone communication,

2

value

[54,110)

but you can’t be sure the e-mail has reached the person.

policy

[111,222)

I would suggest a notice that says, “If you don’t acknowledge receipt of this e-mail, we’ll start phoning you.”

value policy reason

FN

4.6.8

4-13 CasiMedicos casimedicos_0_34 Support

1

Premise

[877,935)

The rest of the symptoms fall perfectly within the picture

Premise

[683,830)

In this question, they give an important piece of information when it comes to suspecting a diagnosis, which is the antitransglutaminase antibodies

4-14 ArgUNSC UNSC_2014_SPV.7154_spch002_UNBriefer_neg support

1

premise

[304,563)

Following close to two weeks of relative calm, starting on 6 April armed individuals and separatist groups began to seize Government buildings and confiscate lethal weapons in Donetsk oblast of Ukraine, including in the cities of Donetsk, Lugansk and Kharkiv.

claim

[100,159)

the situation regarding Ukraine has seriously deteriorated.

CasiMedicos ArgUNSC premise claim train unsplit_author_release ArgUNSC AAEC CDCP

4.6.9 CDCP

v0 v0 κR=1 TP ≤5% F1 39.81

4-15 CDCP v0

κR

TP

F1

F1

Platt

0.5

41.00%

29.95

39.81

Platt

0.75

100.00%

4.29

39.81

Platt

1.0

100.00%

4.04

39.81

Platt

1.25

100.00%

2.53

39.81

raw

1.0

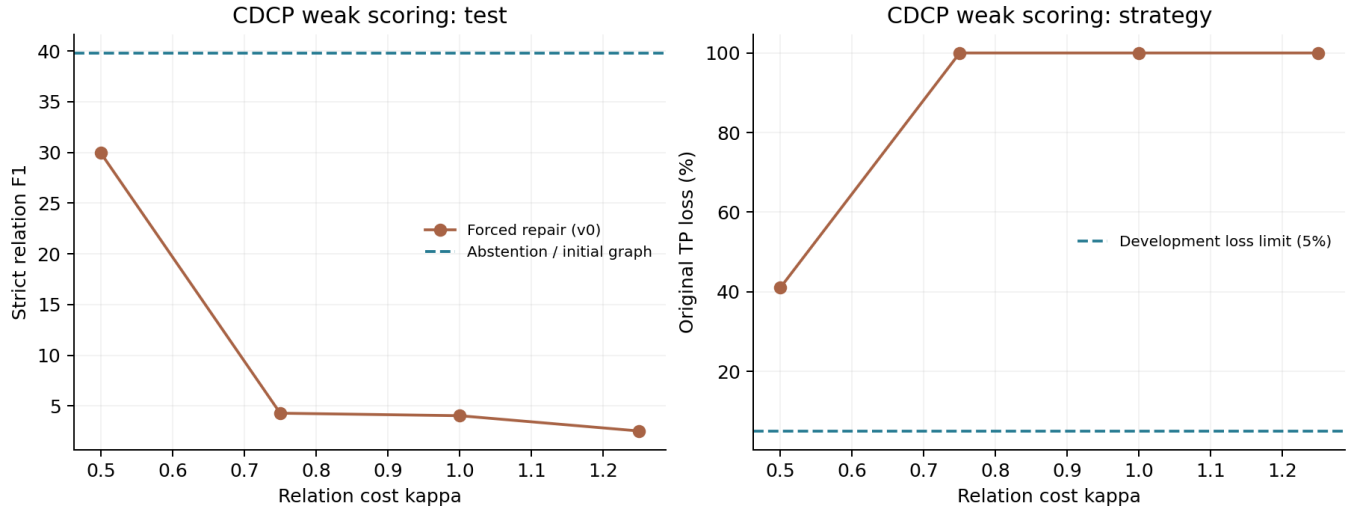
100.00%

6.46

39.81

κR 0.5 v0 TP 41% κR=1 δ=0.20 60/115 TP κR=0.5 0/115
AAEC 3/561 δ 0.20

AUROC ECE TP v0



4-4 CDCP

4.7 □□□□

F1

F1

5.1 继承基线与外部参照

T0/I0 2-2 E UniASA Kawarada TextToText G

5-1

I0

T1

I1

CPU

5.2 四格结果与交互

5-2

AAEC	F1
AAEC	F1
CDCP	F1
CDCP	F1
T0I0	
84.18 ± 0.79	
65.18 ± 3.43	
75.99 ± 1.61	
38.76 ± 3.37	
T1I0	
86.73 ± 0.67	
68.81 ± 2.05	
79.92 ± 1.51	
43.65 ± 3.53	
T0I1	
85.27 ± 0.50	
69.90 ± 2.83	
78.39 ± 1.34	
46.72 ± 2.42	
T1I1	
87.60 ± 0.44	
73.29 ± 1.76	
81.91 ± 1.22	
52.02 ± 2.59	

/pp

95% CI

AAEC

T1I0 – T0I0

3.63

[1.78, 5.60]

1.85 / 4.45 / 4.57

AAEC

T1I1 – T1I0

4.48

[3.43, 5.54]

3.88 / 4.90 / 4.66

AAEC

T0I1 – T0I0

4.72

[3.58, 5.85]

3.91 / 5.19 / 5.05

AAEC

T1I1 – T0I0

8.10

[5.80, 10.47]

5.73 / 9.35 / 9.23

AAEC

Interaction – 0

-0.24

[-0.55, 0.06]

-0.03 / -0.29 / -0.39

CDCP

T1I0 – T0I0

4.89

[2.91, 7.23]

5.95 / 4.37 / 4.37

CDCP

T1I1 – T1I0

8.37

[6.11, 10.94]

7.14 / 8.20 / 9.77

CDCP

T0I1 – T0I0

7.97

[5.84, 10.32]

7.33 / 7.29 / 9.27

CDCP

T1I1 – T0I0

13.26

[10.31, 16.01]

13.09 / 12.56 / 14.14

CDCP

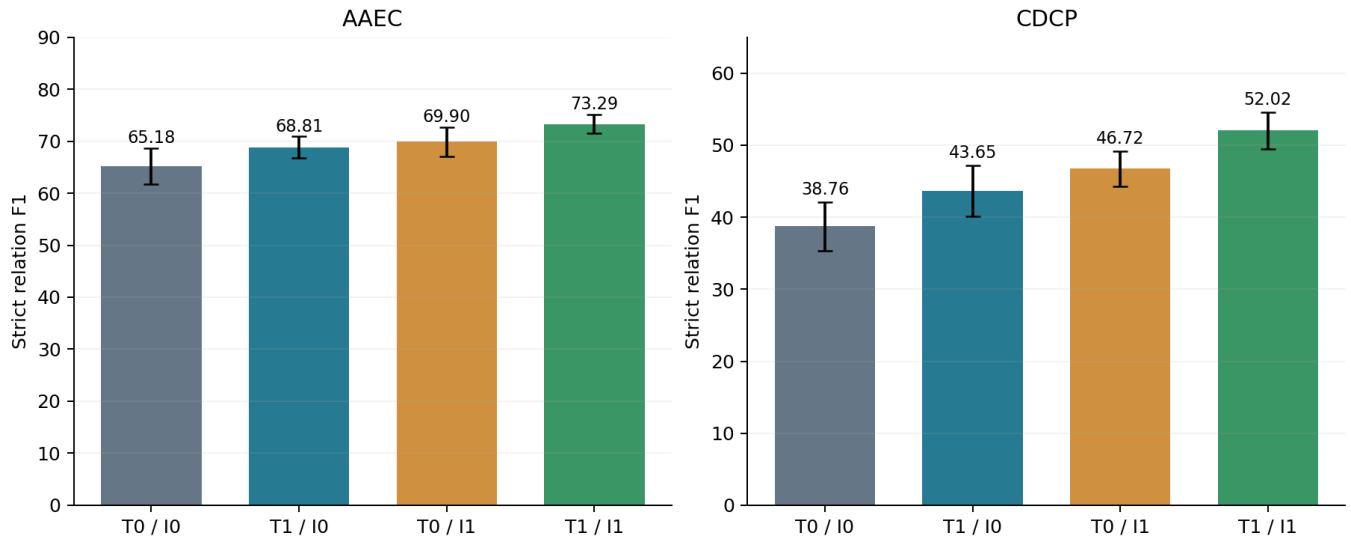
Interaction – 0

0.40

[-0.70, 1.54]

-0.19 / 0.90 / 0.50

T0/I0 AAEC 8.10 pp CDCP 13.26 pp (T1/I1−T1/I0)−(T0/I1−T0/I0)



5-1 F1

5.3

5-4

TP

TP

TP

AAEC

1582

1

93

36

1

0

9

1704

CDCP

293

2

35

26

3

0

1

358

“ ” T1/I0 T0/I1 T0/I0 TP “ ” “ ” T1/I1 TP TP

– – = TP

5-5a

—

TP

AAEC

T0I0

2427

1582

707

138

AAEC

T1I0

2427

1670

608

149

AAEC

T0I1

2427

1616

685

126

AAEC

T1I1

2427

1704

588

135

CDCP

T0I0

756

293

340

123

CDCP

T1I0

756

330

288

138

CDCP

T0I1

756

320

322

114

CDCP

T1I1

756

358

272

126

5-5b

T1/I1

TP

AAEC

2427

1704

588

0

0

135

0

0

CDCP

756

358

272

2

0

124

0

0

0

“ ” “ ”

5.4 CDCP □□

□□□

5-6 CDCP

TP		
FP		
FN		
	F1	± SD
1		
T0I0		
257		
397		
397		
39.30 ± 2.31		
1		
T1I0		

291

363

363

44.50 ± 2.79

1

T0I1

284

250

370

47.79 ± 1.60

1

T1I1

318

222

336

53.26 ± 1.98

2

T0I0

25

35

35

41.67 ± 10.41

2

T1I0

27

33

33

45.00 ± 5.00

2

T0I1

25

23

35

45.98 ± 9.34

2

T1I1

27

22

33

49.45 ± 2.87

3

T0I0

9

9

9

50.00 ± 16.67

3

T1I0

10

8

8

55.56 ± 25.46

3

T0I1

9

5

9

56.36 ± 18.45

3

T1I1

10

5

8

59.90 ± 23.49

≥4

T0I0

2

22

22

8.33 ± 7.22

≥4

T1I0

2

22

22

8.33 ± 7.22

≥4

T0I1

2

15

22

9.52 ± 8.25

≥4

T1I1

3

13

21

15.02 ± 0.63

218 20 6 8 252 34 5-6 TP
G

5.5 本章小结

6.1 研究总结

AAEC CDCP F1 73.29 52.02 8.10 13.26 pp Holm

I1

6.2 适用条件与局限

F1 AAEC 4/240 CDCP 2/450 CDCP v0

CDCP CasiMedicos 69 ArgUNSC

6.3 后续研究

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A

A.1 □□□□□□□□□□

		[start,end)	Python	Unicode		token			offset mapping
AAEC	CDCP	10,986	AAEC	3	2	CDCP	19	0	24

A.2 □□□□□□□□

A-1 SHA-256

SHA-256

data/corpus.jsonl

7dff0832622e39125fa52fa18cb9dfda169af5b2e33b71d8c03f9410af167a5d

results/split_manifest.json

e6b1b675e0e58754cd0aa561c484cb3138ee076f85f30b0950967f11d6e947b0

results/training_roles.json

17bb1997a7a8fae47cd0c39143ca7fb27f9b155cf958887af0f7b068ddb4afb5

results/registration_complete.json

67abb64feadec549b6c32f87d6ed465c3ec413627f20beb3455c3b09af940e6e

results/training_counts.jsonl

17c2625f6b60cb1829df469469b8caf89f49f21c2bd0c228ec052f6d141a445a

results/frozen_snapshots.jsonl

a798175e85aec6dc7a293b5d0a4d235a722ae8d4812495916dc038adea9c7c7c

results/prediction_graphs.jsonl

f3dac306e3c631e986aef24e7f55c77e1a2367c8c8b7b9db46d39ff1ed40a3a0

dataset document seed method uploaded_manifest
download_manifest rt-qwen25- - -s -e5 OOF

A.3 补充语料范围

CasiMedicos ArgUNSC split CasiMedicos 69 ArgUNSC
144 unsplit_author_release AAEC CDCP

A.4 □ 认与可选设置

A-2

κ δ

κC=1 κR AAEC=1 CDCP=0.5 δ=0.20

κR=1 δ TP ≤5% κR=0.5 δ F1 δ

n≤2500

7 ×2 micro-F1 2000 Holm α=0.05

q=sigmoid(a·logit(rawq)+b) κ

A-3 Platt

n

a

b

AAEC

17

C

1071

1.144899

1.060200

AAEC

17

R

4554

1.114010

0.537484

AAEC

29

C

1092

1.191433

1.166175

AAEC

29

R

4608

1.099364

0.517876

AAEC

43

C

1090

1.082817

0.976387

AAEC

43

R

4956

1.001752

0.199744

CDCP

17

C

1231

0.988990

0.532677

CDCP

17

R

3764

0.819074

-0.837470

CDCP

29

C

1189

1.039569

0.515777

CDCP

29

R

3714

1.017828

-0.335367

CDCP

43

C

1187

0.988395

0.527141

CDCP

43

R

3494

0.875053

-0.595704

A.5 □□□□□□□

RT-PREREG-01 rt-v1.0-frozen D0 OOF JSON

protocol_and_source_hash_freeze

D0 + 0

training_and_augmentation

D0 + 3600

OOF_and_calibration

D0 + 172800

strategy_selection

D0 + 259200

test_labels_unsealed

D0 + 388800

CDCP_failure_condition_sensitivity_registered

D0 + 432000

NoPair CompPair T1/I0

SHA-256

SHA-256

AAEC

NoPair

38a7254c9fd0c301d887328973566a9f0c51f6e81f89ceb42677fbcba38b6ee8

eeab9726b1bd0a6e792f90ece16702b9f79aba3732627fea37750d6c73afd800

AAEC

CompPair

38a7254c9fd0c301d887328973566a9f0c51f6e81f89ceb42677fbcba38b6ee8

eeab9726b1bd0a6e792f90ece16702b9f79aba3732627fea37750d6c73afd800

AAEC

T1I0

38a7254c9fd0c301d887328973566a9f0c51f6e81f89ceb42677fbcba38b6ee8

eeab9726b1bd0a6e792f90ece16702b9f79aba3732627fea37750d6c73afd800

CDCP

NoPair

64c85c779e1766a0d133ffebe9b8866ed2c33bc2bbc570edda850d3f31ae253f

31f11653481964213dc74d921abad0fd263484354279963a483a60d220861341

CDCP

CompPair

64c85c779e1766a0d133ffebe9b8866ed2c33bc2bbc570edda850d3f31ae253f

31f11653481964213dc74d921abad0fd263484354279963a483a60d220861341

CDCP

T1I0

64c85c779e1766a0d133ffebe9b8866ed2c33bc2bbc570edda850d3f31ae253f

31f11653481964213dc74d921abad0fd263484354279963a483a60d220861341

A.6 原文序列化与无损对象身份

AAEC essay001

{"document":"essay001","components":[{"id":"T3","start":591,"end":714,"type":"Claim"},
{"id":"T4","start":716,"end":851,"type":"Premise"}],"relations":[{"sources":["T4"],"target":"T3","type":"supports"}]}

3-1 AAEC stance

supports attacks

B

B.1 数据与模型身份

C0 R0 V/P qC/qR κ OOF

B.2 评分与接受轨迹

G0

CDCP

B.3 回退与预算

≤24 96 ≤24 96 32 3 4 incumbent

G0 token

B.4 实现一致性验收

B.4.1

8970 6900 3131

B.4.2

B-1

coordinates

10986

10986

0

24 partial-token spans retain residual character offsets

answer_mask

288

288

0

max hidden difference under answer replacement=0

gradient_path

36

36

0

shared adapter norm C=.0118 R=.0064; base delta=0; specialized branch align gradient=0

normalization

48

48

0

empty group=0; duplicate-view normalization max error=1.1e-7; last batch correct

candidate_protection

720

720

0

protected objects lost=0; 12 resource overflow cases return G0

MILP_feasibility

240

240

0

small-graph exhaustive objective max error=1.2e-8

timeout

60

60

0

timeout with incumbent=38; without incumbent=22; legal output=60

commit_rollback

40

40

0

injected commit failure restores full graph/hash; compute cost retained

interrupt_resume

24

24

0

resume terminal semantic hash matches uninterrupted run

native_object_counts

320

320

0

full source-set identity; proper subset not a hit

score_missing

60

60

0

NaN/absent scored as missing; no default low score

tie_break

240

240

0

fixed lexicographic tie order completed

24 LoRA “ ”

B.4.3

12 G0 60 38 incumbent 22 incumbent 40 24 60
NaN 240 gap 0

B.5 案例统一记录

ID split SHA-256 TP/FP/FN CPU

B.6 固定评分性质的证明

aX(o)=2qX(o)−κX ID

0.8 1.0 0.2
F1

κ F1

TP F1

C

C.1 已有工作

MCR-BR[34] LoRA MLP JS

C.2 材料可得性

C.3 记录与版本

CPU SHA-256 final_manifest

D CDCP

D.1 弱评分版 □ 的回退

v0 F1 39.81 κ TP 4-15 v1 v0→v1

D.2 主版本的策略选择

D-1

δ

TP

TP

F1

AAEC

0.01

561

3

0.53%

72.03

AAEC

0.05

561

3

0.53%

72.03

AAEC

0.1

561

3

0.53%

72.03

AAEC

0.2

561

3

0.53%

72.03

CDCP

0.01

115

0

0.00%

54.67

CDCP

0.05

115

0

0.00%

54.67

CDCP

0.1

115

0

0.00%

54.67

CDCP

0.2

115

0

0.00%

54.67

CDCP κR=1 0.5 5% F1 F1

D.3 实际 CPU 抽取与后处理

CPU-BIO-SGD SGD log_loss (alpha=10^{-5} 12) (alpha=10^{-4} 20) 17 29 43 max(20,8×) 80 150

— — AAEC CDCP CPU

D.4 解释范围

CPU essay098 CDCP CasiMedicos ArgUNSC

E MCR-BR

MCR-BR[34] “±” A

E.1 ICIC 原版 AAEC 域内结果记录

E-1 ICIC AAEC

Comp-F1

Support-F1

Attack-F1

Relation-F1

FLAN T5-XXL

80.15

62.34

48.72

61.19

GPT-4-turbo 20-shot

55.51

29.86

18.45

28.38

RoBERTa-large MTL

76.43±0.55

54.21±0.52

41.83±0.67

52.15±0.54

LLM Single-Pass

77.82±0.51

58.43±0.45

44.21±0.62

56.87±0.48

LLM w/ Constrained Decoding

78.54±0.46

59.12±0.42

45.33±0.58

57.65±0.45

LLM w/ DAG Post-process

78.91±0.49

59.45±0.48

45.87±0.61

58.12±0.47

LLM Multi-Sample (LogProb)

79.45±0.38

60.12±0.41

46.58±0.55

58.94±0.42

LLM w/ Self-Refine

79.82±0.45

60.75±0.43

47.12±0.56

59.63±0.45

MCR-BR ($K=1$, w/o correction)

81.24±0.44

61.87±0.39

48.34±0.58

60.73±0.41

MCR-BR (Verifier-only Rerank, $K=3$)

82.18±0.40

63.15±0.36

49.82±0.52

62.21±0.38

MCR-BR (Full, $K=3$)

83.71±0.42

65.92±0.38

51.47±0.54

64.83±0.38

ICIC Table 1 AAEC [34] “MCR-BR”

E.2 ICIC □□ CDCP □□ □□□□

E-2 ICIC CDCP

	Comp-F1	Reason-F1	Evidence-F1	Relation-F1
FLAN T5-XXL	72.68	35.24	28.41	33.96
RoBERTa-large MTL	68.52±0.58	31.45±0.61	25.72±0.73	30.23±0.64
LLM Single-Pass	71.35±0.48	33.82±0.52	27.15±0.68	32.64±0.55
LLM w/ DAG Post-process	72.14±0.50	34.65±0.49		

28.02±0.63

33.51±0.52

LLM w/ Self-Refine

72.95±0.47

35.58±0.48

28.94±0.60

34.42±0.50

MCR-BR ($K=1$, w/o correction)

74.56±0.45

36.94±0.46

30.27±0.58

35.78±0.48

MCR-BR (Verifier-only Rerank, $K=3$)

75.12±0.41

37.85±0.44

31.42±0.55

36.64±0.45

MCR-BR (Full, $K=3$)

76.45±0.43

39.12±0.45

32.86±0.57

38.17±0.46

ICIC Table 1 CDCP [34]

E.3 ICIC □□□□

□□□□

E-3 ICIC

Comp-F1

Relation-F1

Δ^{Comp}

Δ^{Rel}

Full Method

83.71±0.42

64.83±0.38

—

—

$K = 1$, w/o correction

81.24±0.44

60.73±0.41

-2.47

-4.10

w/o Claim-specific Adapter

81.96±0.48

63.08±0.44

-1.75

-1.75

w/o Cue-aware Prompting

82.34±0.45

62.87±0.40

-1.37

-1.96

w/o Multi-Objective

82.89±0.43

63.54±0.39

-0.82

-1.29

w/o Validity Signal E_{val}

81.56±0.46

61.92±0.43

-2.15

-2.91

ICIC Table 2[34]

E.4 ICIC 原版验证器替换记录

E-4 ICIC

Comp-F1

Relation-F1

Top-1

Qwen2.5-7B

83.71

64.83

100.0%

42.5%

Llama-3-8B

83.71

64.55

94.2%

43.8%

ICIC Table 3[34]

E.5 ICIC 原版 Oracle 记录

E-5 ICIC Oracle

Comp-F1

Relation-F1

Δ^{Rel}

83.71±0.42

64.83±0.38

—

Candidate Oracle $K=3$

85.12±0.36

67.24±0.34

+2.41

Condition Oracle

—

78.42±0.35

+13.59

Output Oracle

100.00

83.35±0.32

+18.52

ICIC Table 5[34]

E.6 ICIC □□□□□□□□□

E-6 ICIC

42.5%

1.8

2.4

LocalRerun

45.2%

RemoveEdge

24.8%

Replace

19.5%

Promote/Demote

10.5%

ID

18.5% → 2.1%

JSON

6.2% → 0.0%

22.4% → 5.8%

E_{val}

75.3% → 97.9%

ICIC Table 6[34] Replace Promote/Demote BR-O BR-C BR-J

E.7 MCR-BR □□□□□□□□

E-7 MCR-BR

AAEC MajorClaim

Claim

AAEC Claim

Claim

AAEC Premise

Premise

CDCP Policy

Claim

CDCP Value

Claim

CDCP Fact

Premise

CDCP Testimony

Premise

CDCP Reference

Premise

AAEC/CDCP

E.8 ICIC □□ AAEC □ CDCP □

□□□

E-8 ICIC AAEC CDCP

Coarse Component-F1

Unlabeled Link-F1

FLAN T5-XXL zero-shot

52.34±0.65

31.25±0.72

LLM Single-Pass zero-shot

54.87±0.58

33.42±0.68

MCR-BR ($K=1$, w/o correction, zero-shot)

57.23±0.52

38.76±0.61

MCR-BR (Full, zero-shot)

60.78±0.48

43.21±0.55

ICIC Table 4[34] Link-F1 F1

F

F.1 □□□□□□□□

F-1 “ ”“ ”“ ” “ ” Precision Recall

$x\ D$

D

$c=(s,e,t)$

ID

$G=(C,R)$

CDCP

A

$\Omega(C)$

D_A

A

A

NONE

NULL

NONE

ϕ

token

G_0

U, V, P

P

P, R, F_1

F1

Precision Recall

D, B

$A = D \cup B$

q_C, q_R

Φ_C, Φ_R, Φ

F1

no-op

F.2 □□□□□□

token $B_x(c)$ offset mapping

h_{end}^{x}

4.3 4.5.2 4.6.1 B.3 Oracle 4.6.3 Oracle SelectRegion BuildCandidates EnumerateQueue ScoreObject EvaluateTransaction Commit EvaluateCounts A B

F.3 □□□□□□□□

F-2

3.3.1

3.6.5

3.3.3 3.4

3.6.2—3.6.4

4.2

4.6.3

4.4.1

4.6.6

4.5.6 4.5.7

4.6.2 4.6.4 4.6.5

ICIC

1.3 1-2 2.5.1

2-2 A.5 H.1

G

G.1 方法选择

UniASA[48] Kawarada TextToText[7] QOS

G.2 原生格式转换

AAEC stance CDCP sources 3

FP FN TP

G-1

AAEC

stance

809

CDCP

3

252 8 ≥4

ID

G.3 控制条件与成本身份

TP/FP/FN 14

G.4 □□□□□□□

□

G-2

F1

F1

F1 17 / 29 / 43

AAEC

UniASA

82.99 ± 0.68

63.12 ± 2.43

65.88 / 62.18 / 61.31

AAEC

TextToText

81.75 ± 0.85

61.64 ± 2.18

64.15 / 60.44 / 60.32

AAEC

T0I0

84.18 ± 0.79

65.18 ± 3.43

68.85 / 64.65 / 62.05

AAEC

T1I1

87.60 ± 0.44

73.29 ± 1.76

74.58 / 74.00 / 71.28

CDCP

UniASA

74.20 ± 1.83

36.51 ± 2.60

36.11 / 39.29 / 34.13

CDCP

TextToText

72.84 ± 2.08

34.79 ± 1.21

34.52 / 36.11 / 33.73

CDCP

T0I0

75.99 ± 1.61

38.76 ± 3.37

38.89 / 42.06 / 35.32

CDCP

T1I1

81.91 ± 1.22

52.02 ± 2.59

51.98 / 54.62 / 49.45

G-3 T1/I1

/pp

95% CI

p

AAEC

T1I1 – UniASA

10.16

[8.01, 12.72]

0.0005

AAEC

T1I1 – TextToText

11.65

[9.37, 14.42]

0.0005

CDCP

T1I1 – UniASA

15.51

[12.44, 18.64]

0.0005

CDCP

T1I1 – TextToText

17.23

[13.85, 20.75]

0.0005

CDCP

H

H.1 训练和一 □ 性评分成本

54 36 OOF 12 5 epoch 10 epoch token GPU token

H-1

GPU 9 ×3

OOF GPU 18

GPU 6

AAEC

83.09

30.82

1.64

CDCP

142.64

53.66

2.65

H.2 分层审核的拒绝与域外检查

H-2

changed

undecided

original_disagreement

AAEC

10

6

4

AAEC

13

10

7

AAEC NONE

10

8

3

AAEC NONE

17

10

7

CDCP

9

6

3

CDCP

19

11

11

CDCP NONE

12

8

3

CDCP NONE

15

6

10

Wilson 95%

AAEC

3870

160

2

1.25%

[0.34, 4.44]

CDCP

915

160

2

1.25%

[0.34, 4.44]

J_A 3-6 Nh

H.3 □□

□□□□

H-4

/

P95/

/

AAEC

500

60

0.14

0.32

3

0

60

AAEC

1000

60

0.33

0.72

3

0

60

AAEC

1500

60

0.62

1.22

3

0

60

AAEC

2000

60

0.95

2.03

3

0

60

AAEC

2500

60

1.35

2.71

3

0

60

AAEC

3000

60

1.90

3.00

3

7

53

AAEC

4000

60

2.60

3.00

3

20

40

CDCP

500

60

0.21

0.49

5

0

60

CDCP

1000

60

0.50

1.09

5

0

60

CDCP

1500

60

0.94

1.85

5

0

60

CDCP

2000

60

1.44

3.09

5

0

60

CDCP

2500

60

2.05

4.12

5

0

60

CDCP

3000

60

2.89

5.00

5

7

53

CDCP

4000

60

3.95

5.00

5

20

40

2500 P95 3000 4000

H.4 □□□

□□□□

H-5

TP

FP

FN

F1

AAEC

T0I0

17

557

252

252

68.85

AAEC

T0I0

29

523

286

286

64.65

AAEC

T0I0

43

502

307

307

62.05

AAEC

T1I0

17

572

237

237

70.70

AAEC

T1I0

29

559

250

250

69.10

AAEC

T1I0

43

539

270

270

66.63

AAEC

T0I1

17

565

179

244

72.76

AAEC

T0I1

29

536

190

273

69.84

AAEC

T0I1

43

515

211

294

67.10

AAEC

T1I1

17

581

168

228

74.58

AAEC

T1I1

29

572

165

237

74.00

AAEC

T1I1

43

551

186

258

71.28

CDCP

T0I0

17

98

154

154

38.89

CDCP

T0I0

29

106

146

146

42.06

CDCP

T0I0

43

89

163

163

35.32

CDCP

T1I0

17

113

139

139

44.84

CDCP

T1I0

29

117

135

135

46.43

CDCP

T1I0

43

100

152

152

39.68

CDCP

T0I1

17

104

94

148

46.22

CDCP

T0I1

29

115

99

137

49.36

CDCP

T0I1

43

101

100

151

44.59

CDCP

T1I1

17

118

84

134

51.98

CDCP

T1I1

29

127

86

125

54.62

CDCP

T1I1

43

113

92

139

49.45

H.5 六项一致性问题的处理

H-6

ECE

q

× ×

Nh π

I0 I1

F1 SD

1-2 4-4 14 Holm

H.6 复算入口

scripts/scenario.py statistics_and_supplements.py validate_and_sensitivity.py make_complete_records.py finalize_records.py real_cases.py build_review_manuscript.py render_artifacts.py export_word.py check_delivery.py CPU Markdown DOCX README final_manifest

I

I.1 □□□
□□□□
□□

I-1

issues

AAEC

test

80

1266

809

0

0

0

AAEC

train

319

4793

3006

0

0

0

CDCP

test

150

1026

252

0

0

0

CDCP

train

580

3901

969

0

0

0

argunsc

unsplit_author_release

144

4560

3009

0

0

36

casimedicos

test

117

1509

684

0

0

70

casimedicos

train

381

4979

2145

0

0

241

casimedicos

validation

55

739

341

0

0

30

CasiMedicos 622 train 434 53 validation 63 8 test 125 8 381 55 117 69
ID issues 341 ArgUNSC 144 36 issues

I.2 案例定位与原始哈希

I-2

split

SHA-256

AAEC

essay001

train

ba75cbadbcaf04c7b0b0b4b4e552958c2a7a880314787f58a24c96eb7fbc17fd

AAEC

essay098

test

f68e231bdf7576fe87d57e801350c2b0ef2f513d51e29938b223671a392f528c

AAEC

essay098

test

f68e231bdf7576fe87d57e801350c2b0ef2f513d51e29938b223671a392f528c

CDCP

00662

train

b69b5e4700be7b0bfe4ac92179bbc906be67f7b6435442e033a3c44906bd58c9

casimedicos

casimedicos_0_34

train

dd0704fe2f47c8bce117779659b7f79957603b67a53f218fffa0506b967e9b63

argunsc

argunsc_UNSC_2014_SPV.7154_spch002_UNBriefer_neg

unsplit_author_release

acc4543591e9260ed99dcc8361cd0e0bd69c39a1436c2d23fdb4dddf7cbcb1c6

text[start:end] essay098 CDCP 00662 CasiMedicos ArgUNSC real_cases.json real_case_counts